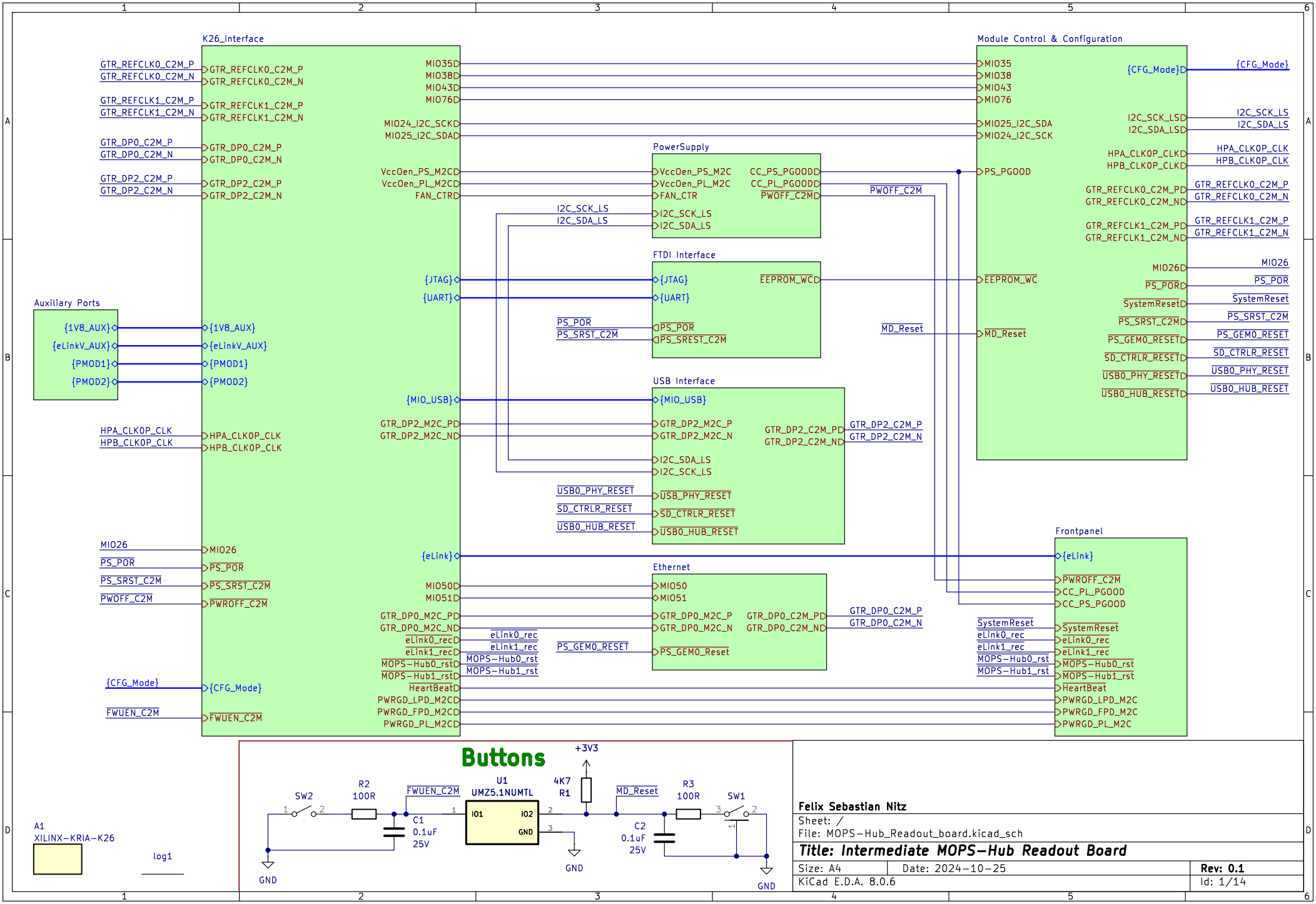
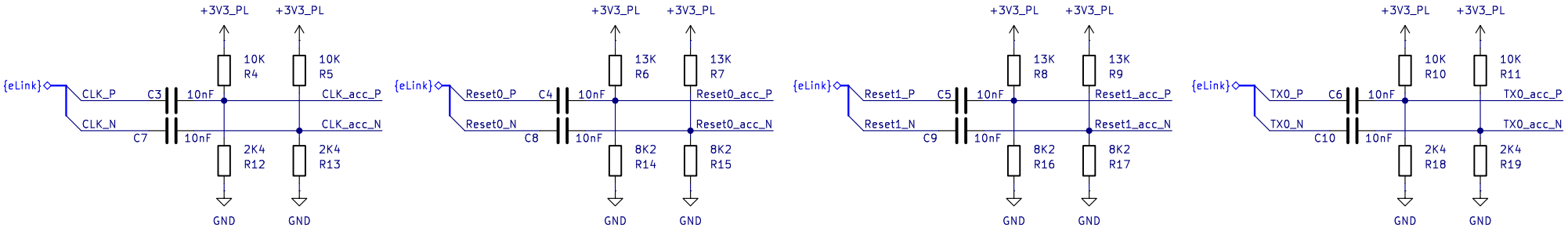
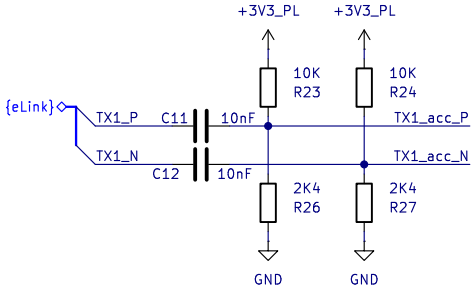
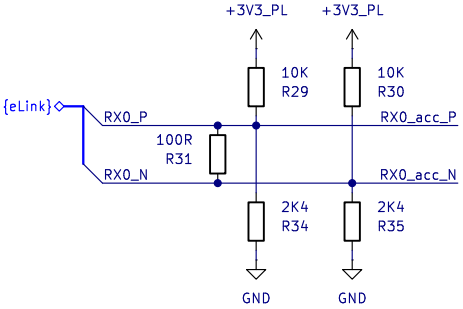
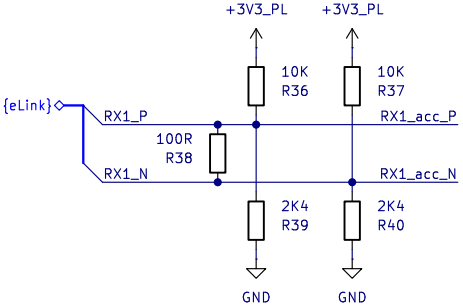

K26_Interface
Module Control & Configuration
PowerSupply
FTDI Interface
USB Interface
Ethernet
Frontpanel
Auxiliary Ports
Buttons
+3V3
U1
UMZ5.1NUMTL
SW2
SW1
R2
100R
R3
100R
4K7
R1
C1
0.1uF
25V
C2
0.1uF
25V
GND
A1
XILINX-KRIA-K26
log1
Felix Sebastian Nitz
Sheet: /
File: MOPS-Hub_Readout_board.kicad_sch
Title: Intermediate MOPS-Hub Readout Board
Size: A4
Date: 2024-10-25
Rev: 0.1
KiCad E.D.A. 8.0.6
Id: 1/14

eLink AC Coupling & Biasing

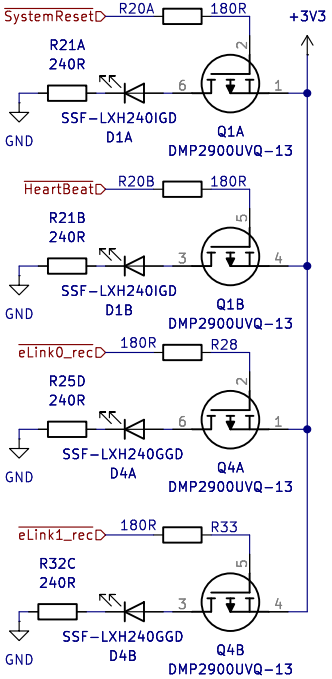


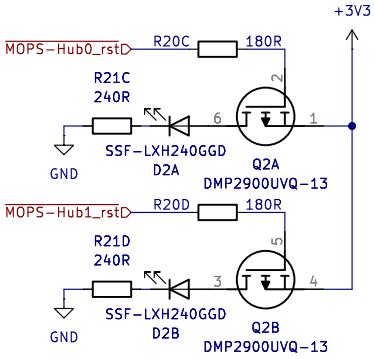




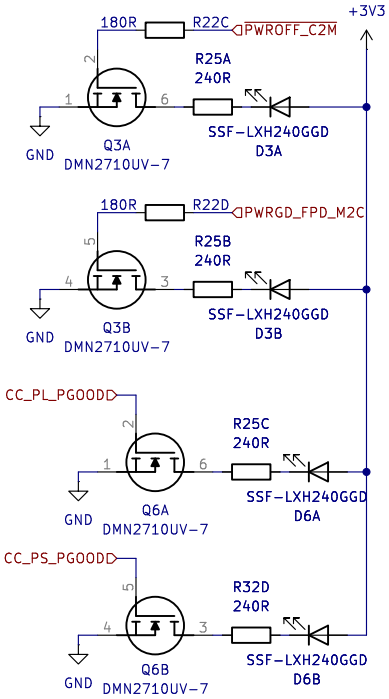


System Status LEDs

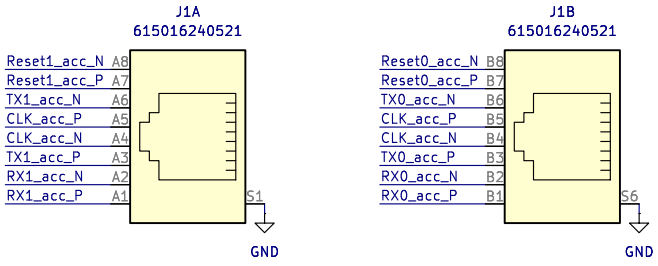




Power Status LEDs



eLink Connectors

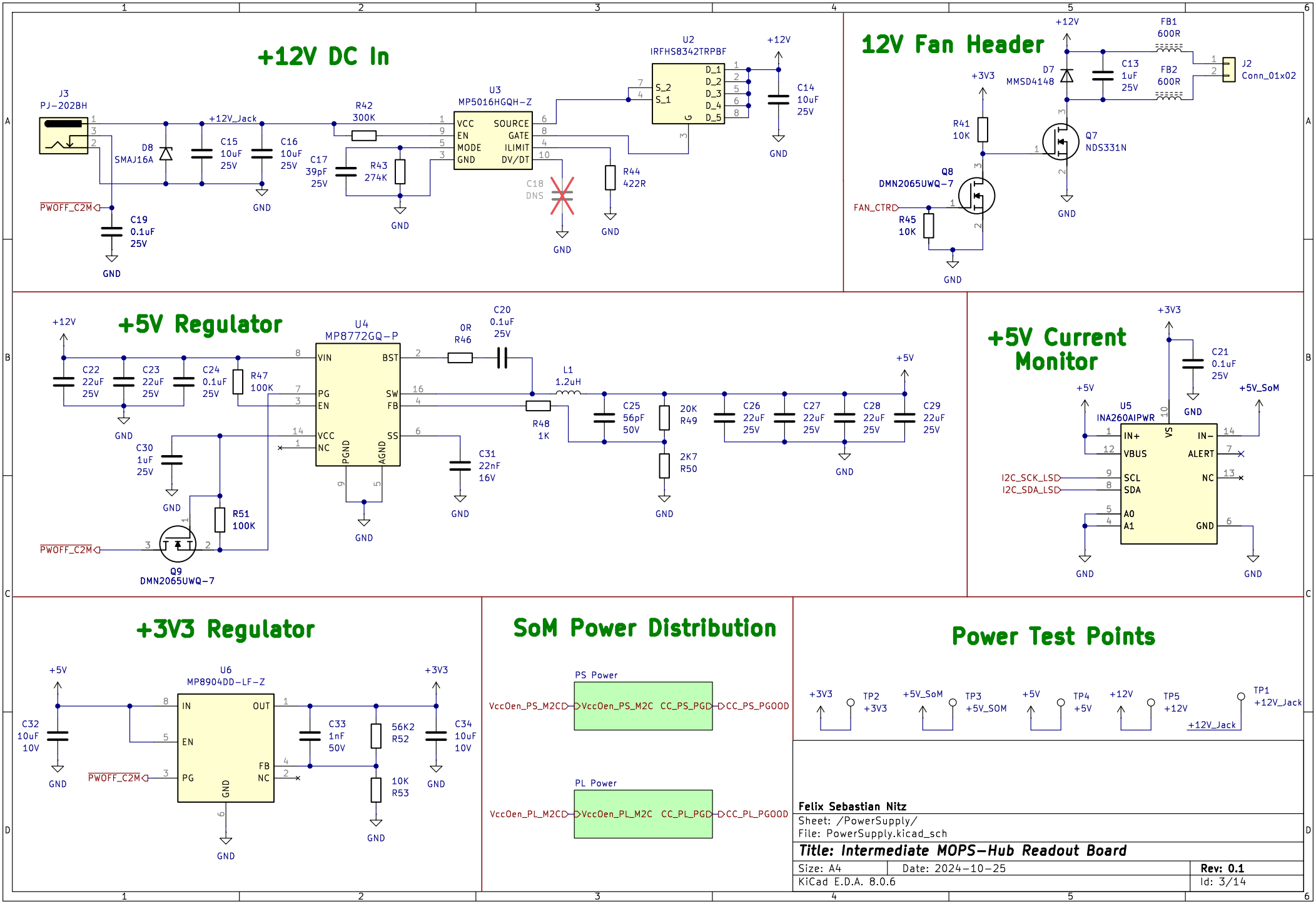


Felix Sebastian Nitz

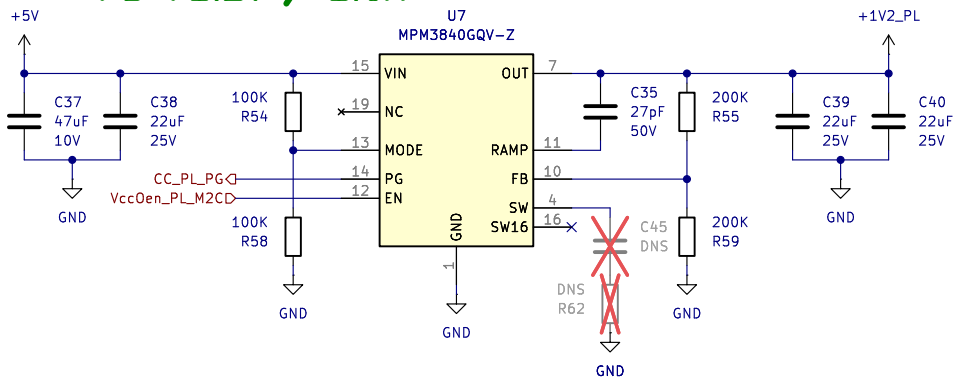
Sheet: /Frontpanel/
File: Frontpanel.kicad_sch

Title: Intermediate MOPS-Hub Readout Board

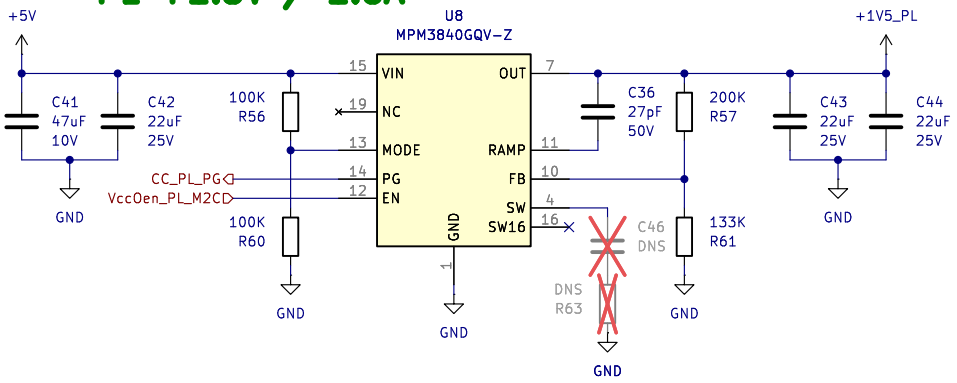
Size: A4	Date: 2024-10-25	**Rev: 0.1**
KiCad E.D.A. 8.0.6		Id: 2/14


+12V DC In
J3
PJ-202BH
D8
SMAJ16A
+12V_Jack
C15
10uF
25V
C16
10uF
25V
R42
300K
U3
MP5016HGQH-Z
VCC
EN
MODE
GND
SOURCE
GATE
ILIMIT
DV/DT
C17
39pF
25V
R43
274K
C18
DNS
R44
422R
U2
IRFHS8342TRPBF
S_2
S_1
G
D_1
D_2
D_3
D_4
D_5
+12V
C14
10uF
25V
PWOFF_C2M
C19
0.1uF
25V
GND
12V Fan Header
+12V
FB1
600R
FB2
600R
J2
Conn_01x02
D7
MMSD4148
C13
1uF
25V
+3V3
R41
10K
Q7
NDS331N
Q8
DMN2065UWQ-7
FAN_CTRL
R45
10K
+5V Regulator
+12V
U4
MP8772GQ-P
VIN
PG
EN
VCC
NC
PGND
AGND
BST
SW
FB
SS
C22
22uF
25V
C23
22uF
25V
C24
0.1uF
25V
R47
100K
C20
0.1uF
25V
0R
R46
L1
1.2uH
R48
1K
C25
56pF
50V
20K
R49
2K7
R50
C26
22uF
25V
C27
22uF
25V
C28
22uF
25V
C29
22uF
25V
+5V
C30
1uF
25V
C31
22nF
16V
R51
100K
Q9
DMN2065UWQ-7
+5V Current Monitor
+3V3
C21
0.1uF
25V
+5V
+5V_SoM
U5
INA260AIPWR
IN+
IN-
VBUS
ALERT
SCL
SDA
NC
A0
A1
VS
I2C_SCK_LSD
I2C_SDA_LSD
+3V3 Regulator
+5V
U6
MP8904DD-LF-Z
IN
OUT
EN
PG
FB
NC
GND
C32
10uF
10V
C33
1nF
50V
56K2
R52
10K
R53
C34
10uF
10V
SoM Power Distribution
PS Power
VccOen_PS_M2C
VccOen_PS_M2C
CC_PS_PGD
CC_PS_PGOOD
PL Power
VccOen_PL_M2C
VccOen_PL_M2C
CC_PL_PGD
CC_PL_PGOOD
Power Test Points
+3V3
TP2
+3V3
+5V_SoM
TP3
+5V_SOM
+5V
TP4
+5V
+12V
TP5
+12V
TP1
+12V_Jack
+12V_Jack
Felix Sebastian Nitz
Sheet: /PowerSupply/
File: PowerSupply.kicad_sch
Title: Intermediate MOPS-Hub Readout Board
Size: A4
Date: 2024-10-25
Rev: 0.1
KiCad E.D.A. 8.0.6
Id: 3/14

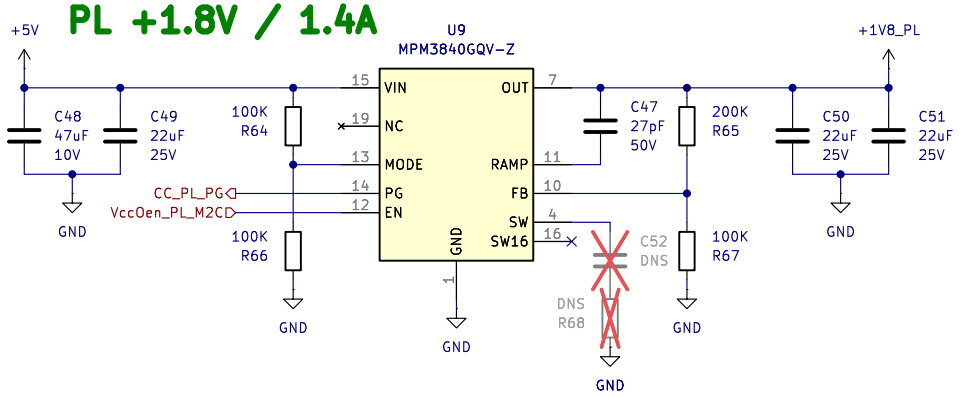
PL +1.2V / 1.8A



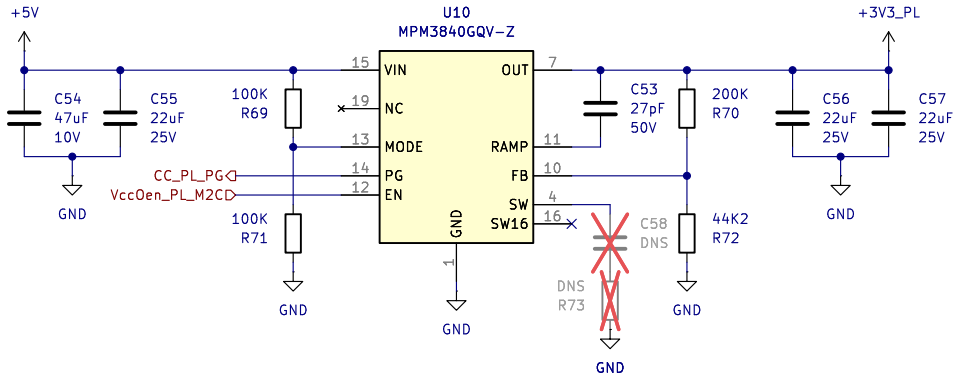
PL +1.5V / 1.8A



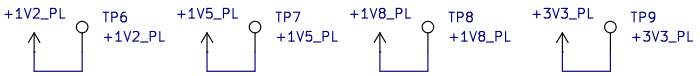
PL +1.8V / 1.4A



PL +3.3V / 3A



Power Test Points



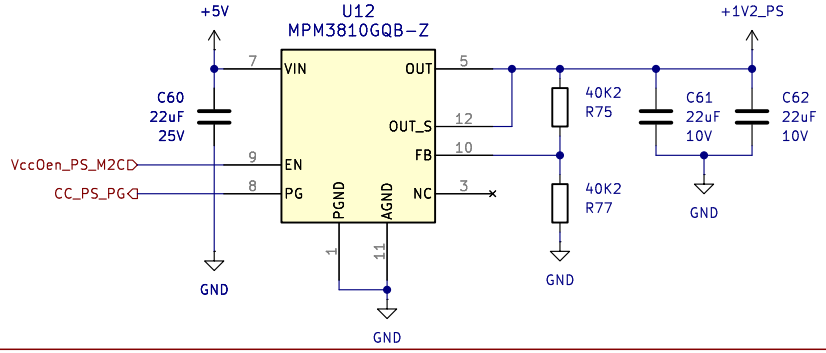
Felix Sebastian Nitz

Sheet: /PowerSupply/PL Power/
File: PlPower.kicad_sch

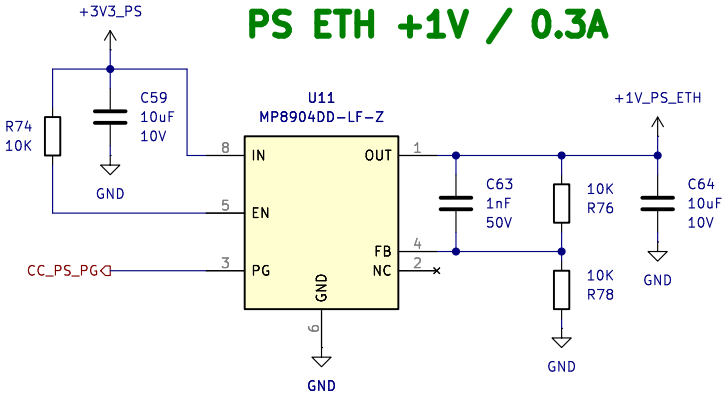
Title: Intermediate MOPS-Hub Readout Board

Size: A4	Date: 2024-10-25	**Rev: 0.1**
KiCad E.D.A. 8.0.6		Id: 4/14

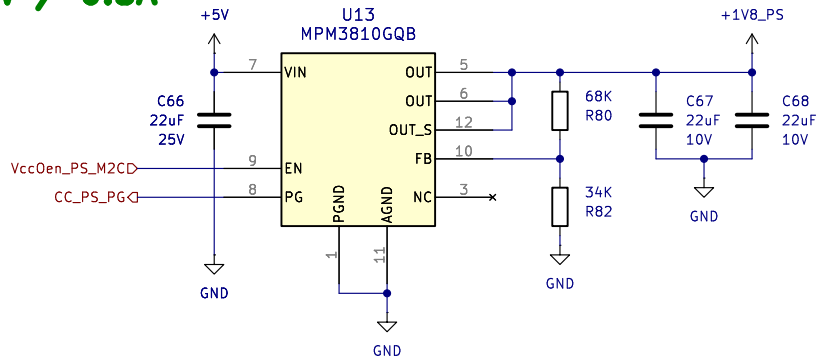
PS +1.2V / 1A



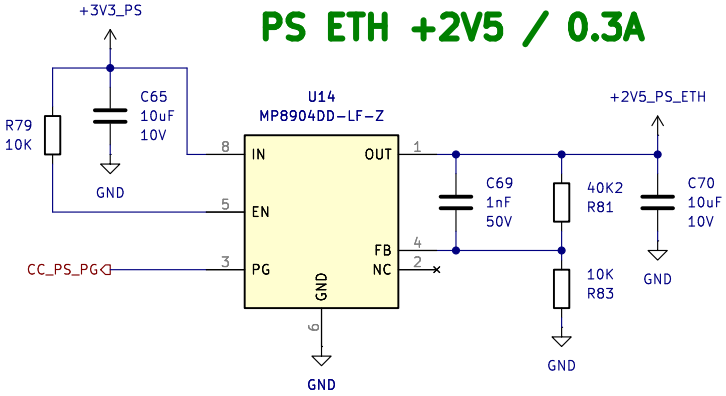
PS ETH +1V / 0.3A



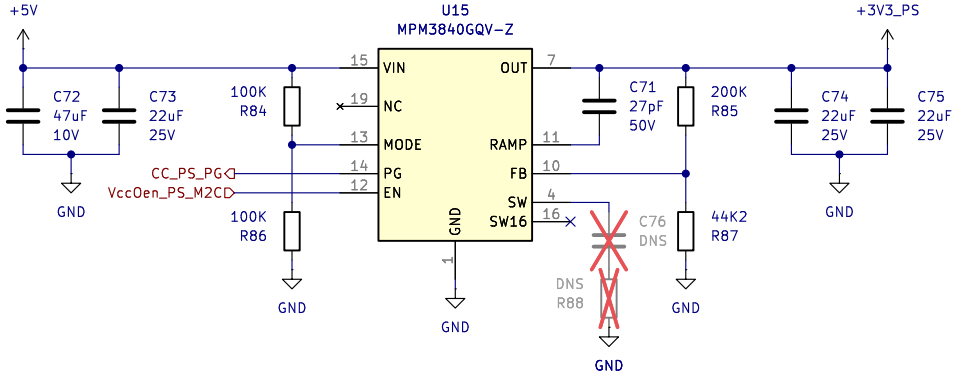
PS +1.8V / 0.5A



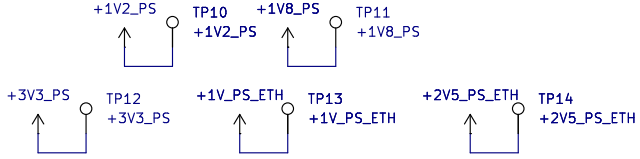
PS ETH +2V5 / 0.3A



PS +3.3V / 2.5A



Power Test Points



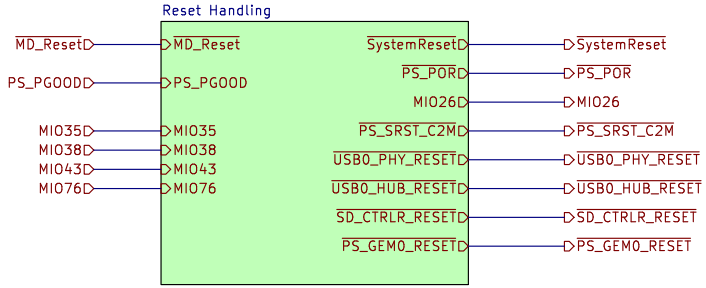
Felix Sebastian Nitz

Sheet: /PowerSupply/PS Power/
File: PsPower.kicad_sch

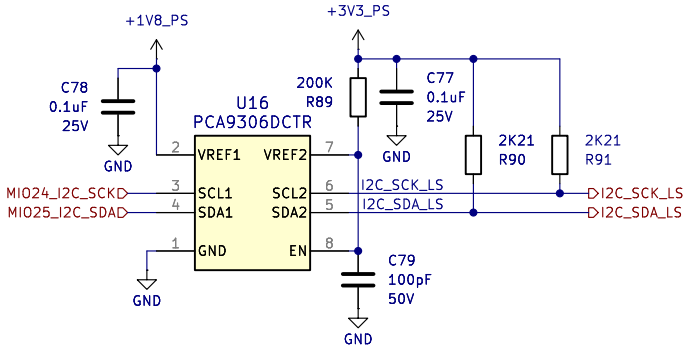
Title: Intermediate MOPS-Hub Readout Board

Size: A4	Date: 2024-10-25	**Rev: 0.1**
KiCad E.D.A. 8.0.6		Id: 5/14

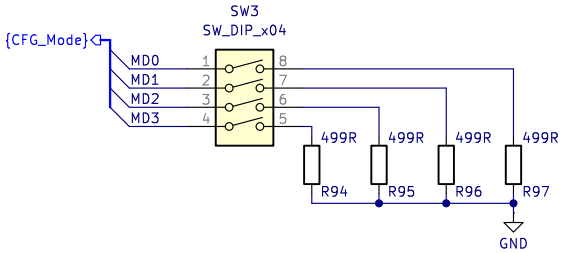
Reset Logic



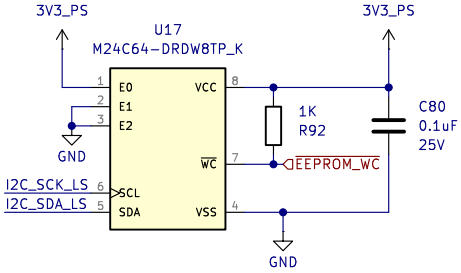
MIO I2C Level Shifter



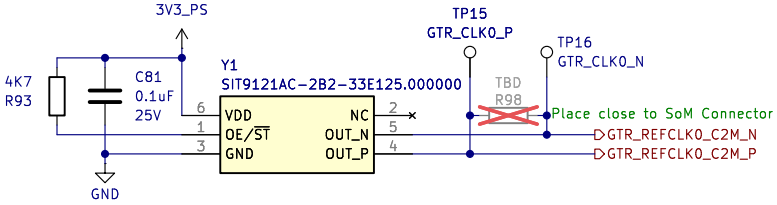
Mode Pins



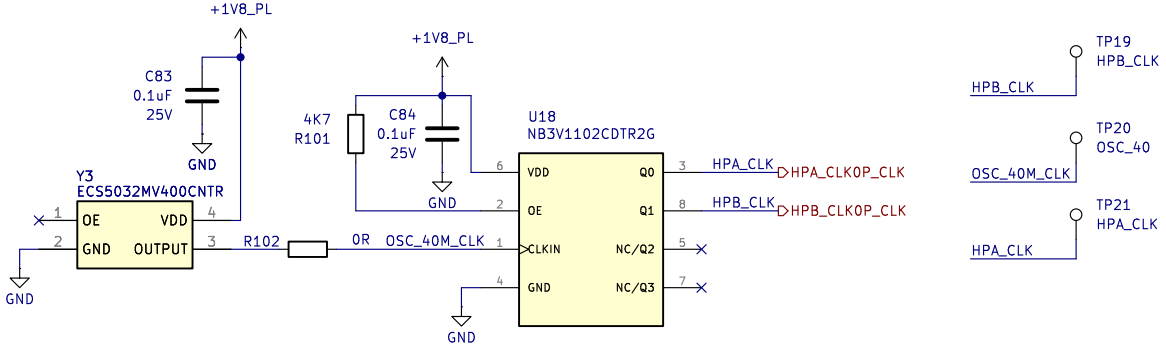
System EEPROM



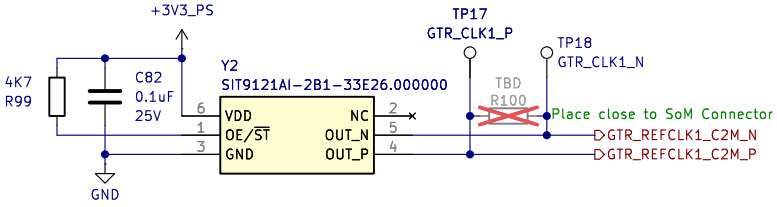
Oscillator: 125MHz for GTR GEM0



User Clock PL / 40MHz LVCMOS



Oscillator: 26MHz for GTR USB



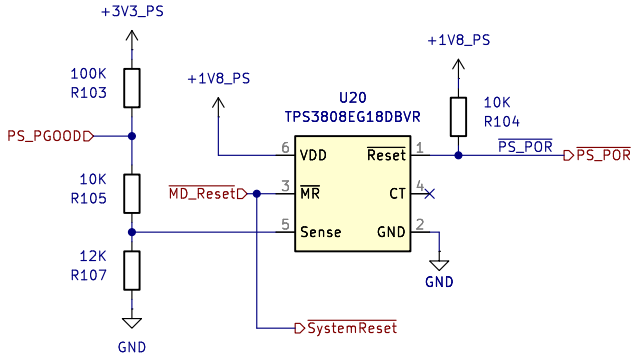
Felix Sebastian Nitz

Sheet: /Module Control & Configuration/
File: ClockNReset.kicad_sch

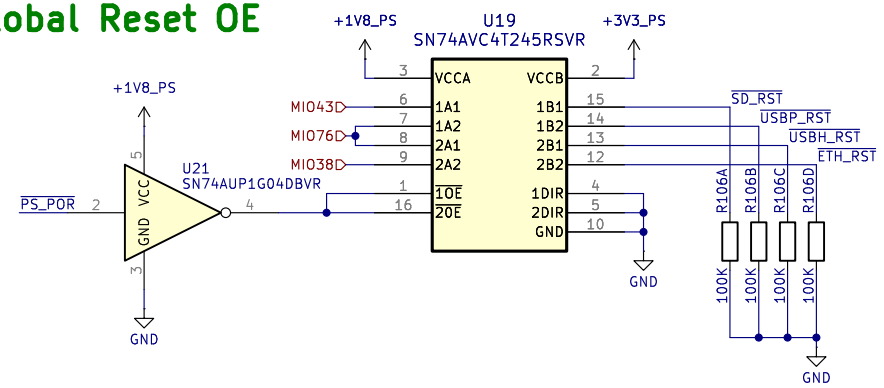
Title: Intermediate MOPS-Hub Readout Board

Size: A4	Date: 2024-10-25	**Rev: 0.1**
KiCad E.D.A. 8.0.6		Id: 6/14

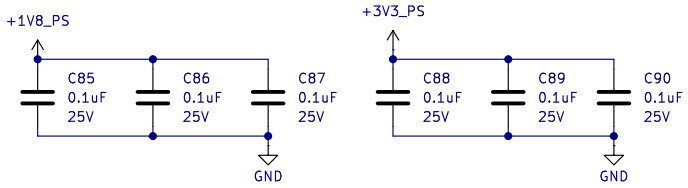
Global Reset



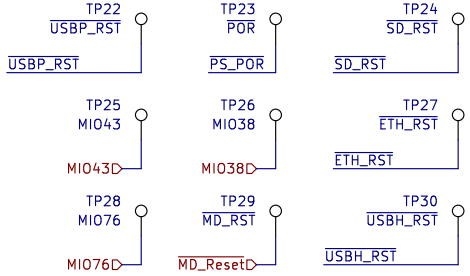
Logic Level Shift With Global Reset OE



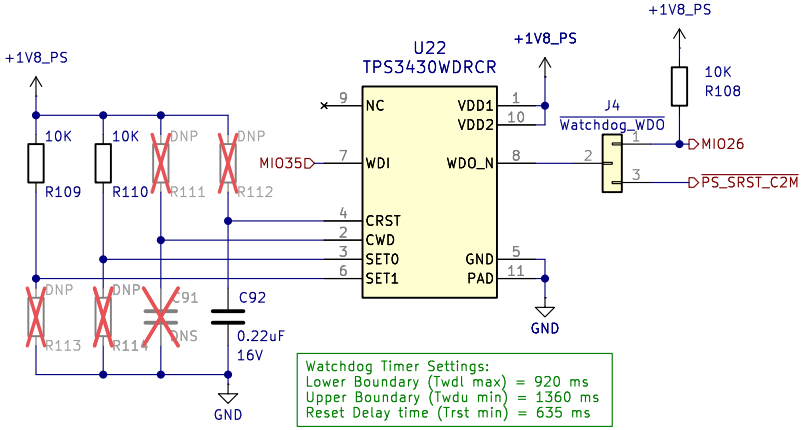
Filtering



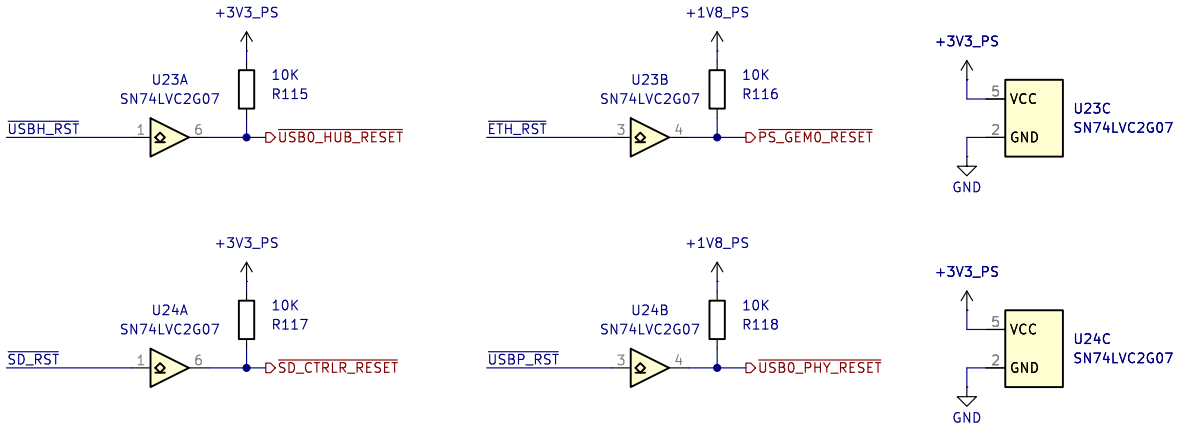
Test Points



Watchdog



Open Drain Reset Signals

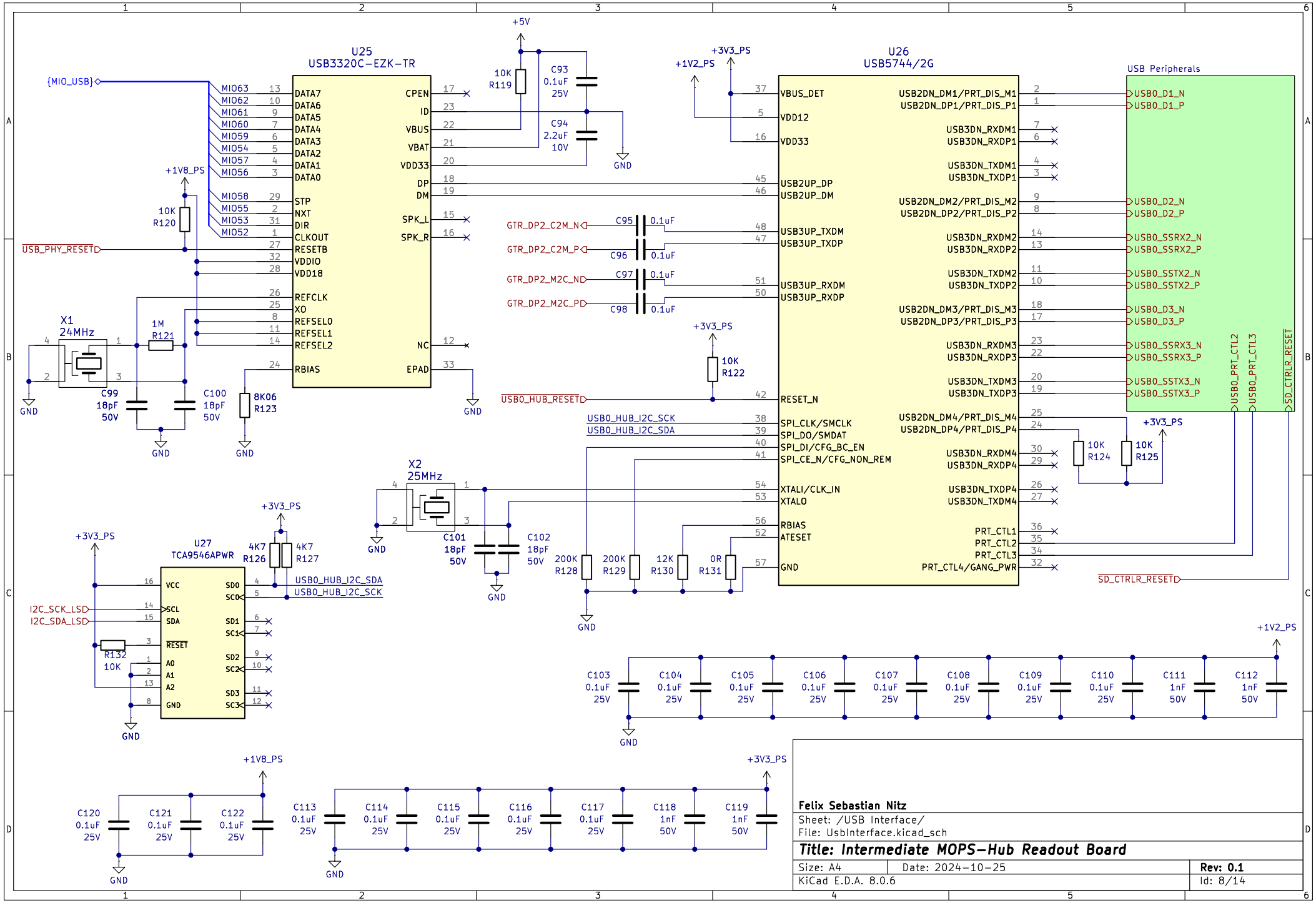


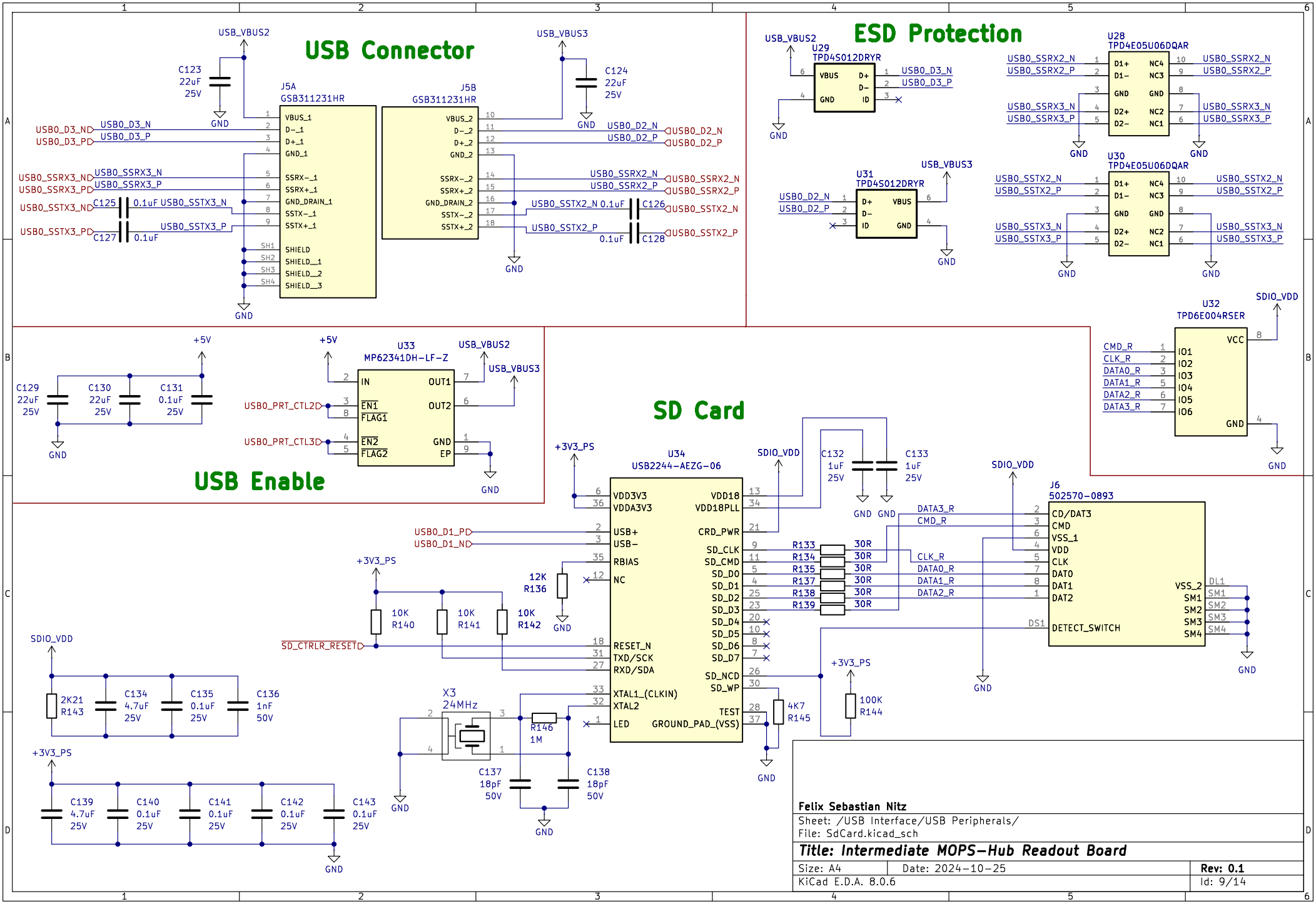
Felix Sebastian Nitz

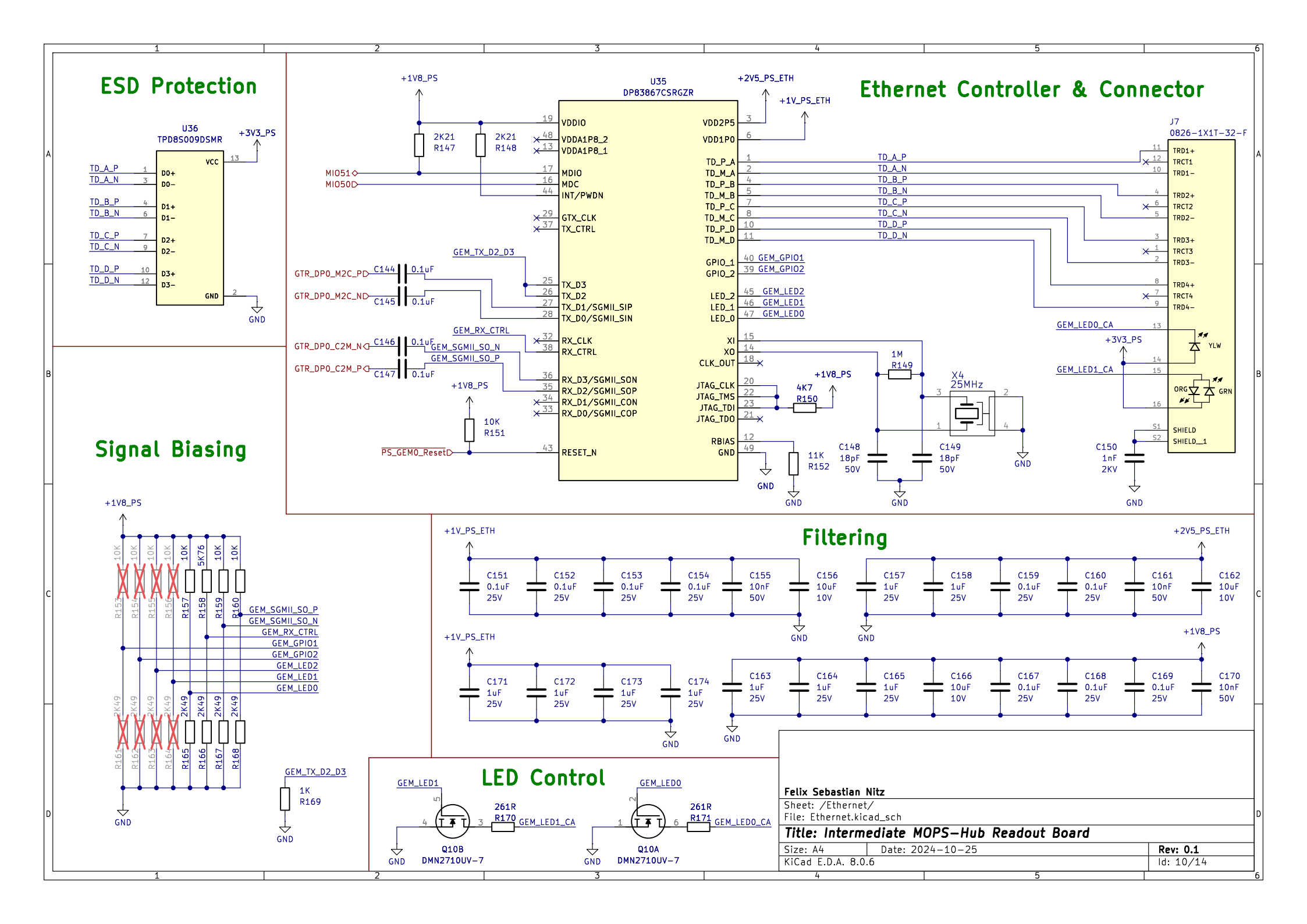
Sheet: /Module Control & Configuration/Reset Handling/
File: Reset.kicad_sch

Title: Intermediate MOPS-Hub Readout Board

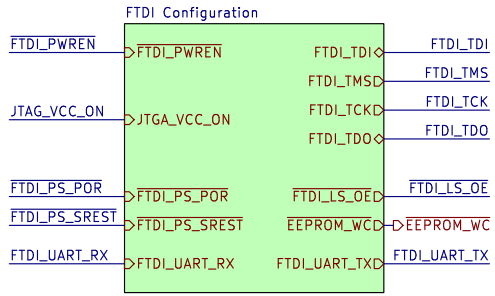
Size: A4	Date: 2024-10-25	**Rev: 0.1**
KiCad E.D.A. 8.0.6		Id: 7/14


U25
USB3320C-EZK-TR
U26
USB5744/2G
USB Peripherals
U27
TCA9546APWR
X1
24MHz
X2
25MHz
{MIO_USB}
USB_PHY_RESET
USB0_HUB_RESET
USB0_HUB_I2C_SCK
USB0_HUB_I2C_SDA
I2C_SCK_LSD
I2C_SDA_LSD
SD_CTRLR_RESET
+5V
+3V3_PS
+1V2_PS
+1V8_PS
GND
Felix Sebastian Nitz
Sheet: /USB Interface/
File: UsbInterface.kicad_sch
Title: Intermediate MOPS-Hub Readout Board
Size: A4
Date: 2024-10-25
Rev: 0.1
KiCad E.D.A. 8.0.6
Id: 8/14

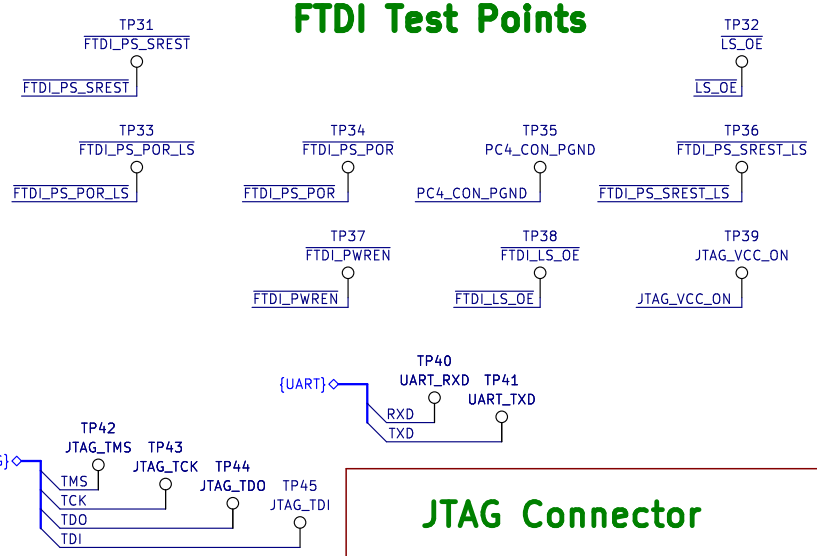

USB Connector
ESD Protection
USB Enable
SD Card
J5A
GSB311231HR
J5B
GSB311231HR
U28
TPD4E05U06DQAR
U29
TPD4S012DRYR
U30
TPD4E05U06DQAR
U31
TPD4S012DRYR
U32
TPD6E004RSER
U33
MP62341DH-LF-Z
U34
USB2244-AEZG-06
J6
502570-0893
X3
24MHz
Felix Sebastian Nitz
Sheet: /USB Interface/USB Peripherals/
File: SdCard.kicad_sch
Title: Intermediate MOPS-Hub Readout Board
Size: A4
Date: 2024-10-25
Rev: 0.1
KiCad E.D.A. 8.0.6
Id: 9/14


ESD Protection
U36
TPD8S009DSMR
+3V3_PS
TD_A_P
TD_A_N
TD_B_P
TD_B_N
TD_C_P
TD_C_N
TD_D_P
TD_D_N
VCC
D0+
D0−
D1+
D1−
D2+
D2−
D3+
D3−
GND
Ethernet Controller & Connector
U35
DP83867CSRGZR
+1V8_PS
+2V5_PS_ETH
+1V_PS_ETH
2K21 R147
2K21 R148
MIO51
MIO50
VDDIO
VDDA1P8_2
VDDA1P8_1
MDIO
MDC
INT/PWDN
GTX_CLK
TX_CTRL
VDD2P5
VDD1P0
TD_P_A
TD_M_A
TD_P_B
TD_M_B
TD_P_C
TD_M_C
TD_P_D
TD_M_D
GPIO_1
GPIO_2
LED_2
LED_1
LED_0
GEM_GPIO1
GEM_GPIO2
GEM_LED2
GEM_LED1
GEM_LED0
GEM_TX_D2_D3
GTR_DP0_M2C_P
GTR_DP0_M2C_N
C144 0.1uF
C145 0.1uF
TX_D3
TX_D2
TX_D1/SGMII_SIP
TX_D0/SGMII_SIN
GEM_RX_CTRL
GTR_DP0_C2M_N
GTR_DP0_C2M_P
C146 0.1uF
C147 0.1uF
GEM_SGMII_SO_N
GEM_SGMII_SO_P
RX_CLK
RX_CTRL
RX_D3/SGMII_SON
RX_D2/SGMII_SOP
RX_D1/SGMII_CON
RX_D0/SGMII_COP
XI
XO
CLK_OUT
JTAG_CLK
JTAG_TMS
JTAG_TDI
JTAG_TDO
4K7 R150
1M R149
X4 25MHz
RBIAS
GND
11K R152
C148 18pF 50V
C149 18pF 50V
PS_GEM0_Reset
10K R151
RESET_N
J7
0826-1X1T-32-F
TRD1+
TRCT1
TRD1−
TRD2+
TRCT2
TRD2−
TRD3+
TRCT3
TRD3−
TRD4+
TRCT4
TRD4−
GEM_LED0_CA
GEM_LED1_CA
YLW
ORG
GRN
SHIELD
SHIELD_1
C150 1nF 2KV
Signal Biasing
+1V8_PS
10K
5K76
2K49
GEM_SGMII_SO_P
GEM_SGMII_SO_N
GEM_RX_CTRL
GEM_GPIO1
GEM_GPIO2
GEM_LED2
GEM_LED1
GEM_LED0
GEM_TX_D2_D3
1K R169
Filtering
C151 0.1uF 25V
C152 0.1uF 25V
C153 0.1uF 25V
C154 0.1uF 25V
C155 10nF 50V
C156 10uF 10V
C157 1uF 25V
C158 1uF 25V
C159 0.1uF 25V
C160 0.1uF 25V
C161 10nF 50V
C162 10uF 10V
C171 1uF 25V
C172 1uF 25V
C173 1uF 25V
C174 1uF 25V
C163 1uF 25V
C164 1uF 25V
C165 1uF 25V
C166 10uF 10V
C167 0.1uF 25V
C168 0.1uF 25V
C169 0.1uF 25V
C170 10nF 50V
LED Control
GEM_LED1
GEM_LED0
261R R170
261R R171
GEM_LED1_CA
GEM_LED0_CA
Q10B DMN2710UV-7
Q10A DMN2710UV-7
Felix Sebastian Nitz
Sheet: /Ethernet/
File: Ethernet.kicad_sch
Title: Intermediate MOPS-Hub Readout Board
Size: A4
Date: 2024-10-25
Rev: 0.1
KiCad E.D.A. 8.0.6
Id: 10/14

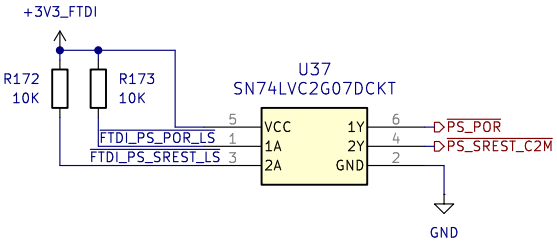
FTDI Configuration Interface



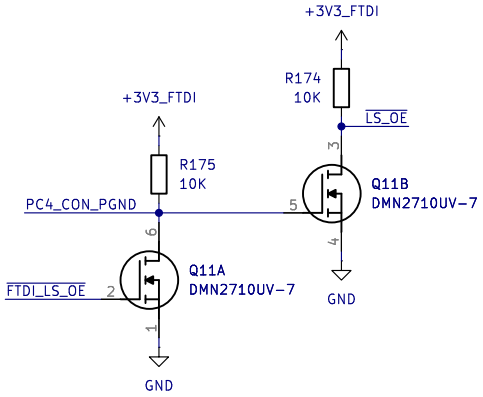
FTDI Test Points



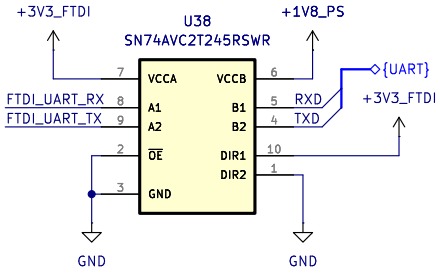
PS POR Drain



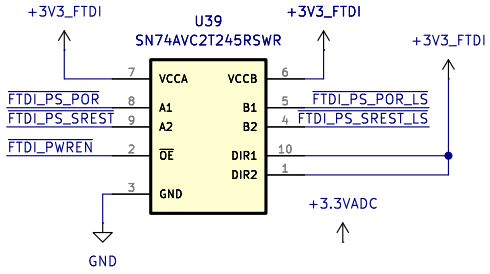
Level Shifter Enable



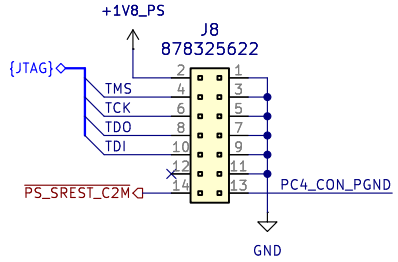
UART Level Shifter



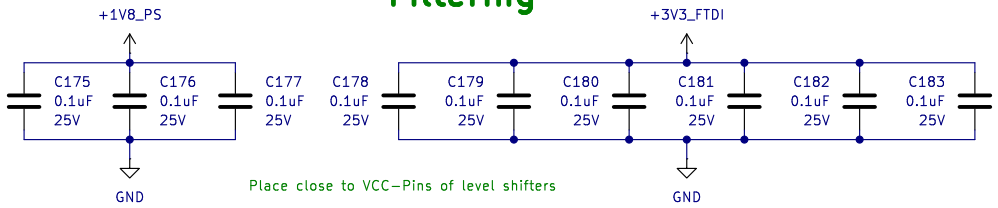
Reset Level Shifter



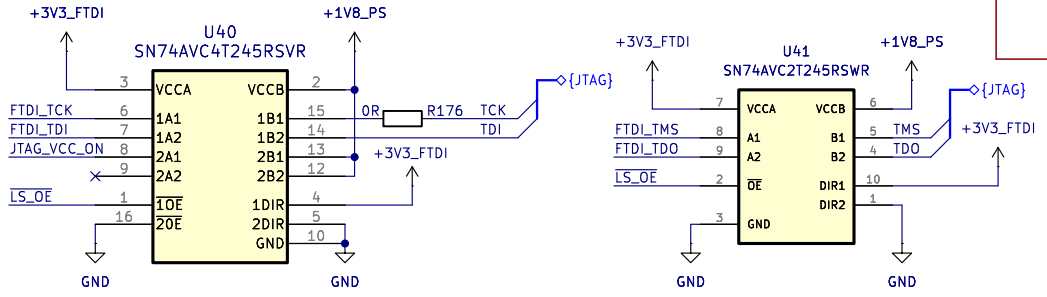
JTAG Connector



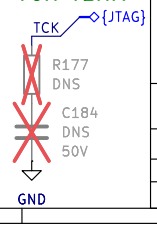
Filtering



JTAG Level Shifter



TCK TERM

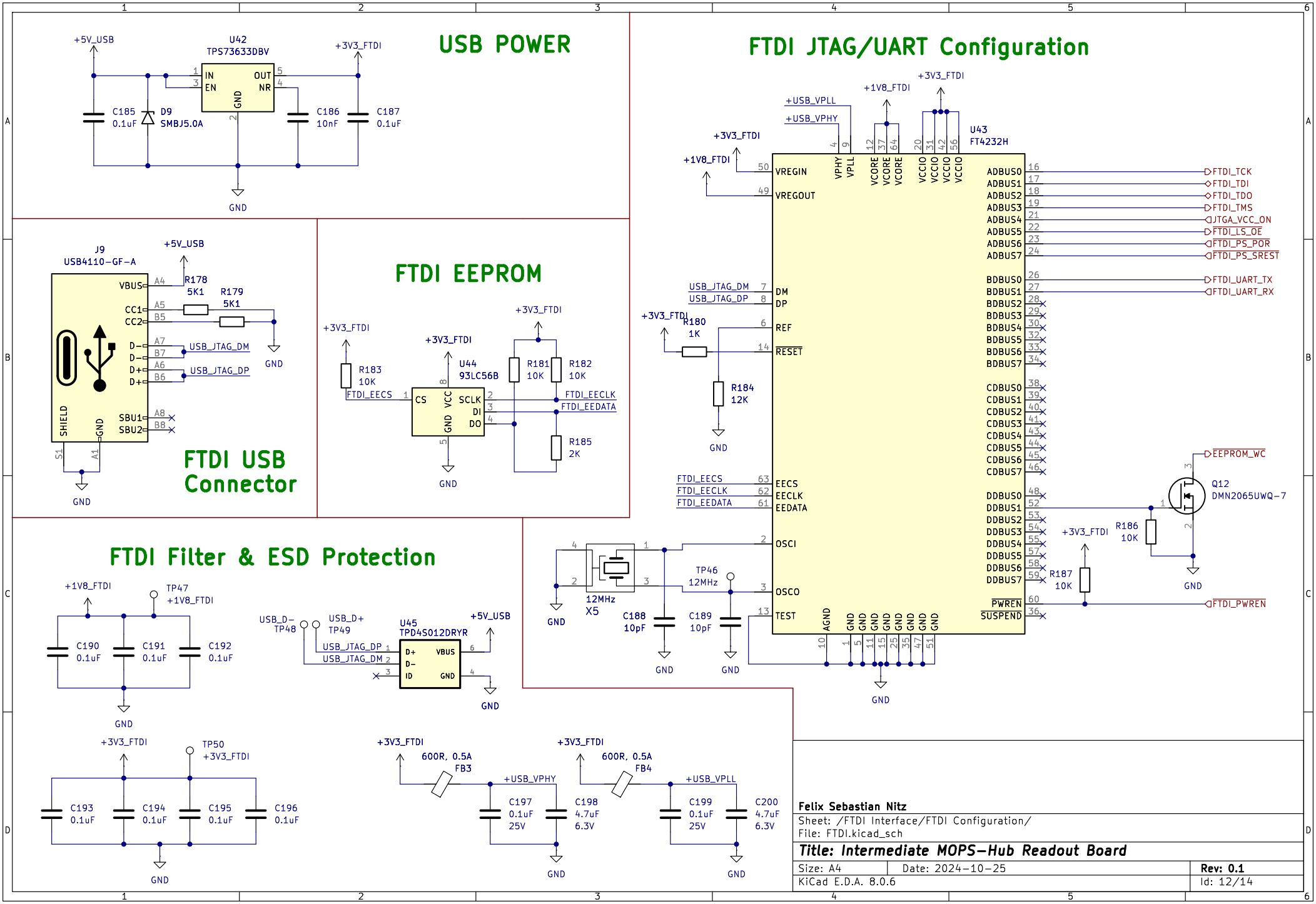


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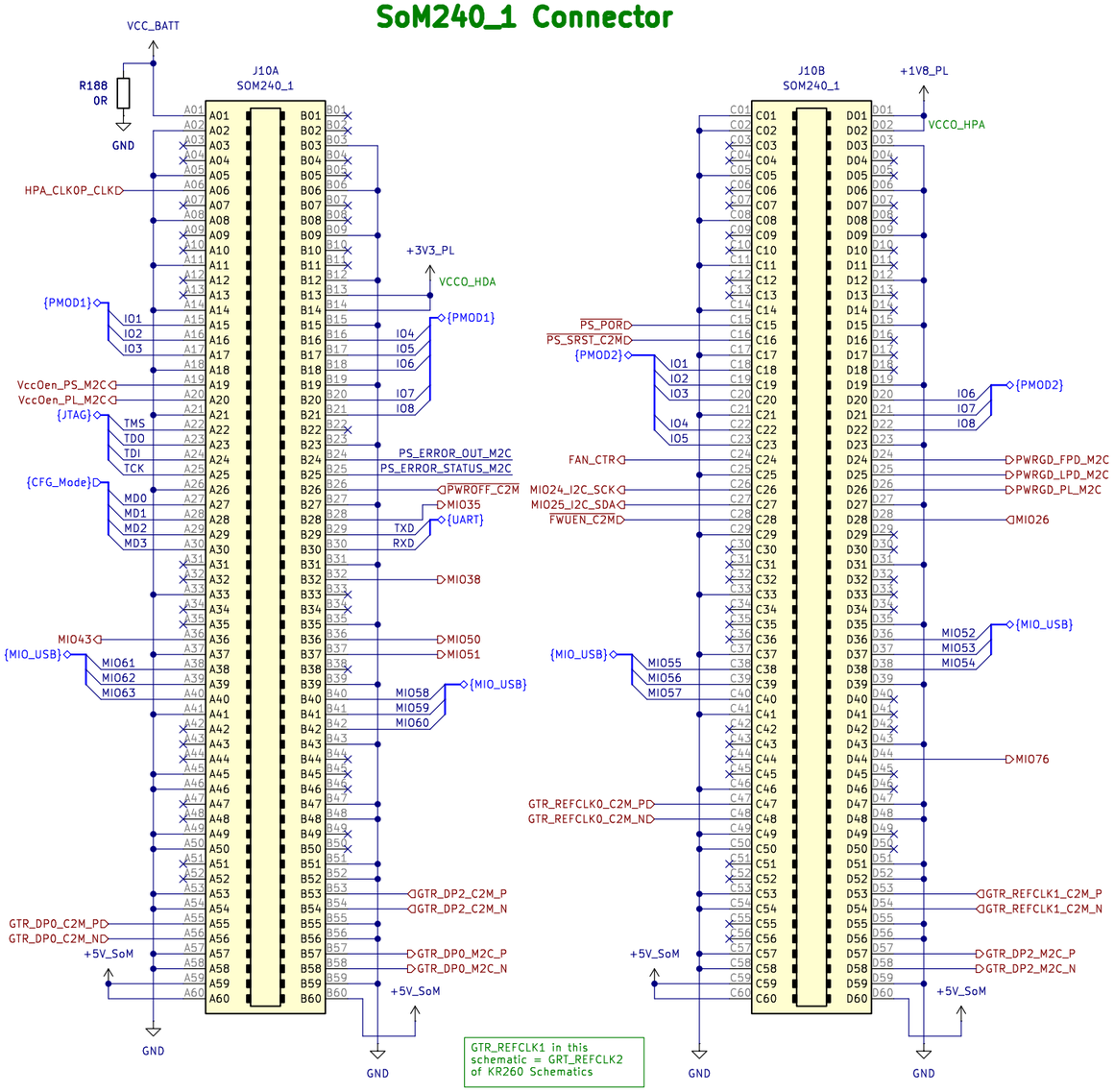
Sheet: /FTDI Interface/
File: FtdiInterface.kicad_sch

Title: Intermediate MOPS-Hub Readout Board

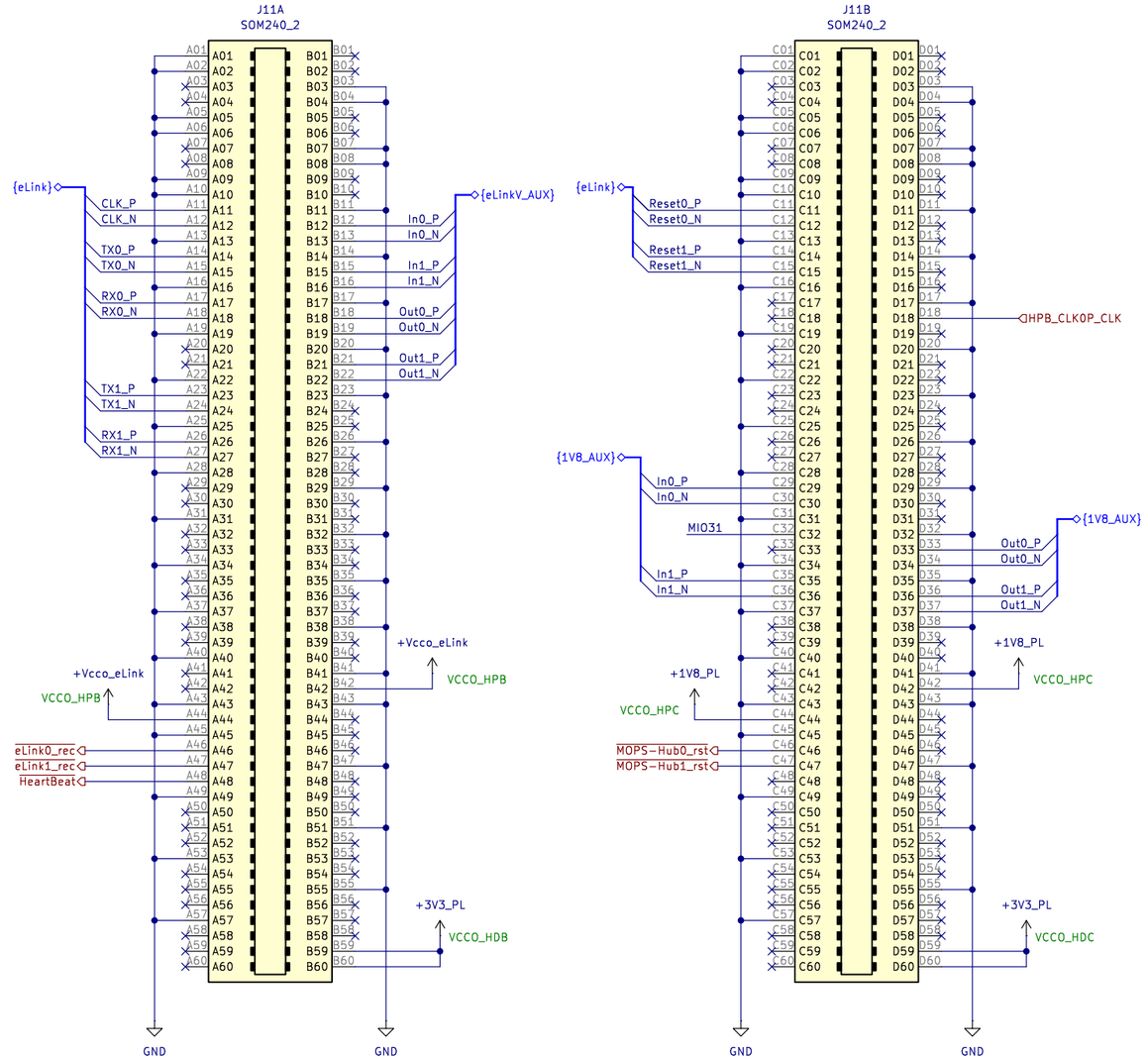
Size: A4	Date: 2024-10-25	Rev: 0.1
KiCad E.D.A. 8.0.6		Id: 11/14


USB POWER
+5V_USB
U42
TPS73633DBV
+3V3_FTDI
IN
EN
OUT
NR
GND
C185
0.1uF
D9
SMBJ5.0A
C186
10nF
C187
0.1uF
GND
FTDI USB Connector
J9
USB4110-GF-A
+5V_USB
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
R178
5K1
R179
5K1
USB_JTAG_DM
USB_JTAG_DP
GND
FTDI EEPROM
+3V3_FTDI
R183
10K
FTDI_EECS
U44
93LC56B
CS
VCC
GND
SCLK
DI
DO
R181
10K
R182
10K
FTDI_EECLK
FTDI_EEDATA
R185
2K
GND
FTDI JTAG/UART Configuration
+3V3_FTDI
+1V8_FTDI
+USB_VPLL
+USB_VPHY
U43
FT4232H
VREGIN
VREGOUT
VPHY
VPLL
VCORE
VCCIO
ADBUS0
ADBUS1
ADBUS2
ADBUS3
ADBUS4
ADBUS5
ADBUS6
ADBUS7
FTDI_TCK
FTDI_TDI
FTDI_TDO
FTDI_TMS
JTGA_VCC_ON
FTDI_LS_OE
FTDI_PS_POR
FTDI_PS_SREST
BDBUS0
BDBUS1
FTDI_UART_TX
FTDI_UART_RX
USB_JTAG_DM
USB_JTAG_DP
DM
DP
REF
RESET
R180
1K
R184
12K
CDBUS0
CDBUS1
CDBUS2
CDBUS3
CDBUS4
CDBUS5
CDBUS6
CDBUS7
EEPROM_WC
Q12
DMN2065UWQ-7
R186
10K
FTDI_EECS
FTDI_EECLK
FTDI_EEDATA
EECS
EECLK
EEDATA
DDBUS0
DDBUS1
DDBUS2
DDBUS3
DDBUS4
DDBUS5
DDBUS6
DDBUS7
R187
10K
FTDI_PWREN
PWREN
SUSPEND
OSCI
OSCO
TEST
12MHz
X5
C188
10pF
C189
10pF
TP46
12MHz
AGND
FTDI Filter & ESD Protection
+1V8_FTDI
TP47
+1V8_FTDI
C190
0.1uF
C191
0.1uF
C192
0.1uF
USB_D-
TP48
USB_D+
TP49
U45
TPD4S012DRYR
+5V_USB
D+
D-
ID
VBUS
GND
USB_JTAG_DP
USB_JTAG_DM
+3V3_FTDI
TP50
+3V3_FTDI
C193
0.1uF
C194
0.1uF
C195
0.1uF
C196
0.1uF
600R, 0.5A
FB3
600R, 0.5A
FB4
+USB_VPHY
+USB_VPLL
C197
0.1uF
25V
C198
4.7uF
6.3V
C199
0.1uF
25V
C200
4.7uF
6.3V
Felix Sebastian Nitz
Sheet: /FTDI Interface/FTDI Configuration/
File: FTDI.kicad_sch
Title: Intermediate MOPS-Hub Readout Board
Size: A4
Date: 2024-10-25
Rev: 0.1
KiCad E.D.A. 8.0.6
Id: 12/14

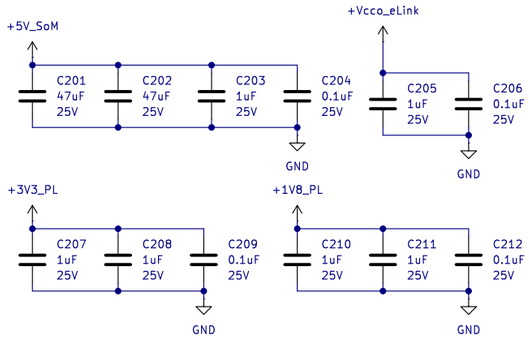
SoM240_1 Connector



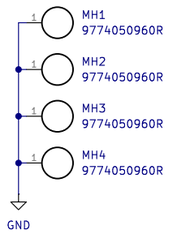
SoM240_2 Connector



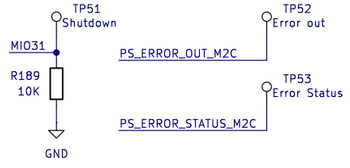
Connector Decoupling



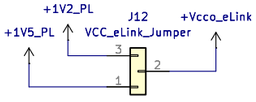
SoM Standoffs



Test Points



VCCO Jumper for eLink Bank

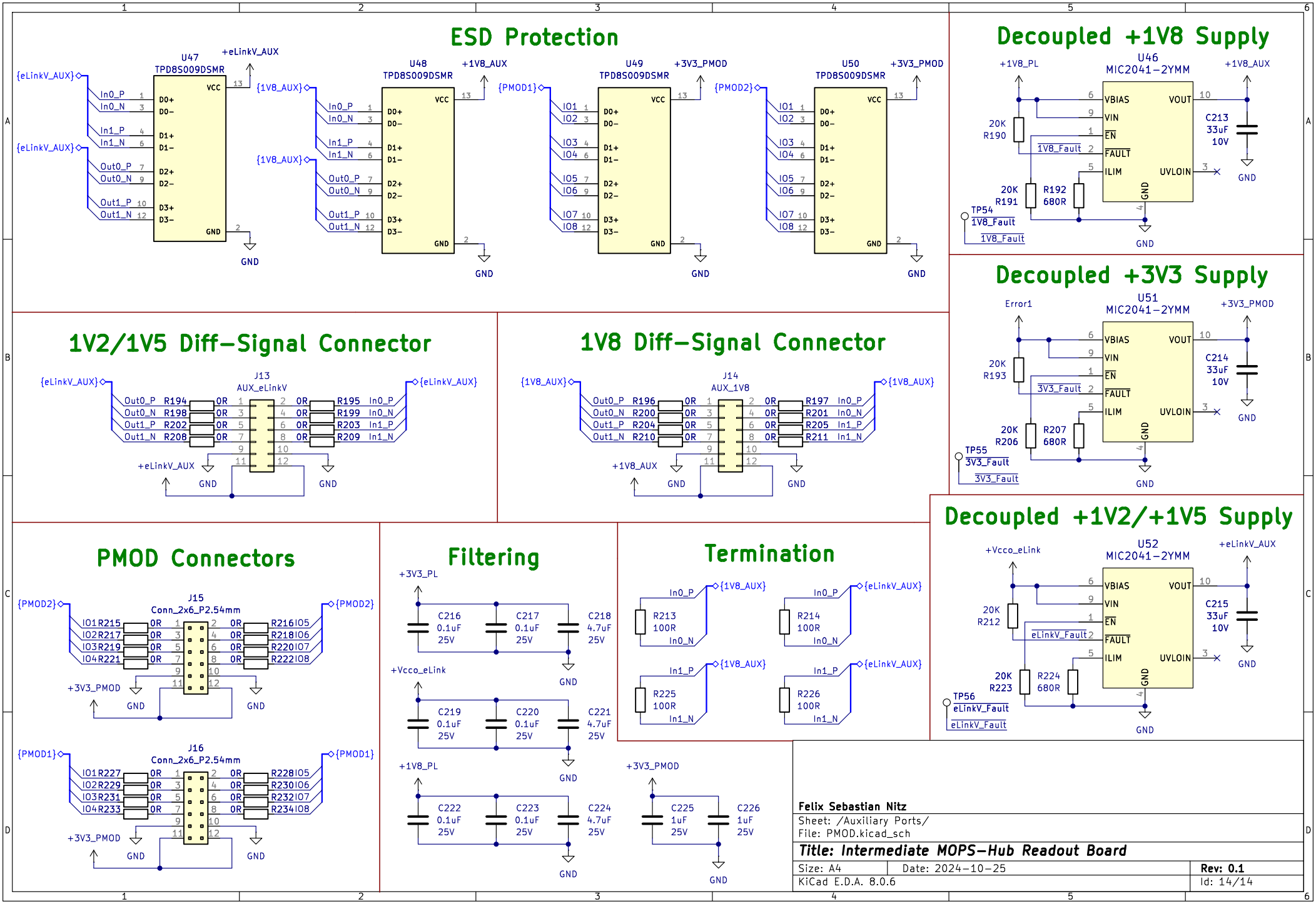


Felix Sebastian Nitz

Sheet: /K26_Interface/
File: K26_Interface.kicad_sch

Title: Intermediate MOPS-Hub Readout Board

Size: A3	Date: 2024-10-25	**Rev: 0.1**
KiCad E.D.A. 8.0.6		Id: 13/14


ESD Protection
U47
TPD8S009DSMR
+eLinkV_AUX
{eLinkV_AUX}
In0_P
In0_N
In1_P
In1_N
Out0_P
Out0_N
Out1_P
Out1_N
D0+
D0−
D1+
D1−
D2+
D2−
D3+
D3−
VCC
GND
U48
TPD8S009DSMR
+1V8_AUX
{1V8_AUX}
U49
TPD8S009DSMR
+3V3_PMOD
{PMOD1}
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
U50
TPD8S009DSMR
+3V3_PMOD
{PMOD2}
Decoupled +1V8 Supply
+1V8_PL
U46
MIC2041−2YMM
+1V8_AUX
20K
R190
VBIAS
VIN
EN
FAULT
ILIM
VOUT
UVLOIN
1V8_Fault
C213
33uF
10V
20K
R191
R192
680R
TP54
1V8_Fault
Decoupled +3V3 Supply
Error1
U51
MIC2041−2YMM
+3V3_PMOD
20K
R193
3V3_Fault
C214
33uF
10V
20K
R206
R207
680R
TP55
3V3_Fault
1V2/1V5 Diff−Signal Connector
J13
AUX_eLinkV
{eLinkV_AUX}
Out0_P
Out0_N
Out1_P
Out1_N
R194
R198
R202
R208
R195
R199
R203
R209
In0_P
In0_N
In1_P
In1_N
0R
+eLinkV_AUX
GND
1V8 Diff−Signal Connector
J14
AUX_1V8
{1V8_AUX}
R196
R200
R204
R210
R197
R201
R205
R211
+1V8_AUX
Decoupled +1V2/+1V5 Supply
+Vcco_eLink
U52
MIC2041−2YMM
+eLinkV_AUX
20K
R212
eLinkV_Fault
C215
33uF
10V
20K
R223
R224
680R
TP56
eLinkV_Fault
PMOD Connectors
J15
Conn_2x6_P2.54mm
{PMOD2}
IO1
IO2
IO3
IO4
R215
R217
R219
R221
R216
R218
R220
R222
IO5
IO6
IO7
IO8
+3V3_PMOD
GND
J16
Conn_2x6_P2.54mm
{PMOD1}
R227
R229
R231
R233
R228
R230
R232
R234
Filtering
+3V3_PL
C216
0.1uF
25V
C217
0.1uF
25V
C218
4.7uF
25V
+Vcco_eLink
C219
0.1uF
25V
C220
0.1uF
25V
C221
4.7uF
25V
+1V8_PL
C222
0.1uF
25V
C223
0.1uF
25V
C224
4.7uF
25V
+3V3_PMOD
C225
1uF
25V
C226
1uF
25V
Termination
In0_P
In0_N
In1_P
In1_N
R213
100R
R214
100R
R225
100R
R226
100R
{1V8_AUX}
{eLinkV_AUX}
Felix Sebastian Nitz
Sheet: /Auxiliary Ports/
File: PMOD.kicad_sch
Title: Intermediate MOPS−Hub Readout Board
Size: A4
Date: 2024−10−25
Rev: 0.1
KiCad E.D.A. 8.0.6
Id: 14/14